

Executive Summary

- ▶ Objective: Predict Falcon 9 first-stage landing success to determine launch price competitiveness.
- ▶ Key Finding: KSC LC-39A is the most successful site with a 76.92% success rate.
- ▶ Booster Performance: The B5 booster version maintains a 100% success record in the dataset.
- ▶ Conclusion: High success probability is linked to technical maturity (Flight Number) and specific site selection.

Introduction

- ▶ SpaceX has disrupted the industry by making rockets reusable.
- ▶ Cost Comparison: \$165M (Traditional) vs. \$62M (SpaceX).
- ▶ Project Goal: Use data science to classify whether a first stage will land successfully.
- ▶ Significance: Enables accurate bidding against SpaceX by predicting their internal costs.

Data Collection & Wrangling

- ▶ API Integration: Targeted SpaceX REST API for launch, rocket, and payload specifics.
- ▶ Web Scraping: Used BeautifulSoup to extract historical launch tables from Wikipedia.
- ▶ Wrangling: Handled missing values in 'Payload Mass' using mean values.
- ▶ Labeling: Converted complex landing outcomes into a binary Class (1 = Success, 0 = Failure).

EDA & Interactive Analytics Methodology

- ▶ Exploratory Analysis: Used Seaborn/Matplotlib for trend identification.
- ▶ Geospatial Analytics: Employed Folium to map sites and proximity to infrastructure (Highways, Railways).
- ▶ Dashboards: Built a Plotly Dash application with dropdowns and range sliders for real-time data exploration.

Predictive Analysis (Machine Learning)

- ▶ Preprocessing: Data was standardized using StandardScaler for algorithm consistency.
- ▶ Algorithms: Logistic Regression, SVM, Decision Tree, and K-Nearest Neighbors (KNN).
- ▶ Tuning: Used GridSearchCV to find optimal hyperparameters (C, kernel, neighbors).
- ▶ Split: 80/20 train-test split with confusion matrix evaluation.

EDA Visualization Results

- ▶ Flight Number vs. Success: Clear upward trend; as experience increases, success becomes the norm.
- ▶ Payload Influence: Heavy payloads do not negatively impact landing success, highlighting Falcon 9's lift capacity.
- ▶ Site Evolution: CCAFS SLC-40 shows a marked improvement in success rates over time.

SQL Analysis Results

- ▶ Total Successes: KSC LC-39A leads with 10 successful landings.
- ▶ Efficiency: KSC LC-39A holds the highest success rate (76.92%).
- ▶ Reliability: SQL query identified 'B5' as the gold standard for booster versions.

Geospatial Analysis Results

- ▶ Coastline Strategy: All launch sites are positioned on the coast to allow for ocean-bound trajectories.
- ▶ Logistics: Sites are within 5-10km of railways/highways but far from populated city centers for safety.
- ▶ Visual Clarity: Marker clusters reveal a higher density of successful landings on the East Coast.

Plotly Dash Dashboard Results

- ▶ Interactivity: Site-specific pie charts allow for localized success analysis.
- ▶ Payload Correlation: Scatter plots show different success densities across booster versions.
- ▶ Utility: Tool allows for 'what-if' scenarios based on payload weight and launch site selection.

Machine Learning Results

- ▶ Model Performance: All models achieved approximately 83.33% accuracy on the test data.
- ▶ Consistency: Logistic Regression, SVM, and KNN showed nearly identical performance.
- ▶ Interpretability: The Decision Tree model provided the clearest 'rules' for success prediction.

Conclusion

- ▶ Landing success is a predictable variable given the right mission parameters.
- ▶ Operational Advantage: SpaceX's learning curve is evident in the data; higher flight numbers equal lower risk.
- ▶ Recommendation: Focus on KSC LC-39A and B5 boosters for maximum reliability.

Innovative Insights



- ▶ The 'Payload Danger Zone': Innovative Insight — Success rates drop to only 22.2% for payloads between 5000-7500kg.
- ▶ Heavy Lift Reliability: Counter-intuitively, payloads over 7500kg have a high 60% success rate.
- ▶ Infrastructure Correlation: Landing success is highly correlated with site-specific ground control maturity.